

Class 14: RNA-Seq Mini-Project

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Background

This lab focuses on using pathway and gene set analysis to interpret large RNA-seq gene lists by identifying enriched biological pathways, processes, and functions. By connecting differentially expressed genes to known annotations, the analysis provides deeper biological insight into the molecular mechanisms underlying the studied condition.

Data Import

```
library(DESeq2)
library(ggplot2)
```

```
metaFile <- "GSE37704_metadata.csv"
countFile <- "GSE37704_featurecounts.csv"
```

```
# Import metadata and take a peak
colData = read.csv(metaFile, row.names=1)
head(colData)
```

```
          condition
SRR493366 control_sirna
SRR493367 control_sirna
SRR493368 control_sirna
SRR493369      hoxa1_kd
SRR493370      hoxa1_kd
SRR493371      hoxa1_kd
```

```
# Import countdata
countData = read.csv(countFile, row.names=1)
head(countData)
```

	length	SRR493366	SRR493367	SRR493368	SRR493369	SRR493370
ENSG00000186092	918	0	0	0	0	0
ENSG00000279928	718	0	0	0	0	0
ENSG00000279457	1982	23	28	29	29	28
ENSG00000278566	939	0	0	0	0	0
ENSG00000273547	939	0	0	0	0	0
ENSG00000187634	3214	124	123	205	207	212
	SRR493371					
ENSG00000186092	0					
ENSG00000279928	0					
ENSG00000279457	46					
ENSG00000278566	0					
ENSG00000273547	0					
ENSG00000187634	258					

Sanity Check

Q1. Complete the code below to remove the troublesome first column from countData.

```
countData <- as.matrix(countData[,-1])
head(countData)
```

	SRR493366	SRR493367	SRR493368	SRR493369	SRR493370	SRR493371
ENSG00000186092	0	0	0	0	0	0
ENSG00000279928	0	0	0	0	0	0
ENSG00000279457	23	28	29	29	28	46
ENSG00000278566	0	0	0	0	0	0
ENSG00000273547	0	0	0	0	0	0
ENSG00000187634	124	123	205	207	212	258

Q2. Complete the code below to filter countData to exclude genes (i.e. rows) where we have 0 read count across all samples (i.e. columns).

Tip: What will rowSums() of countData return and how could you use it in this context?

```
countData <- countData[rowSums(countData) > 0, ]
head(countData)
```

	SRR493366	SRR493367	SRR493368	SRR493369	SRR493370	SRR493371
ENSG00000279457	23	28	29	29	28	46
ENSG00000187634	124	123	205	207	212	258
ENSG00000188976	1637	1831	2383	1226	1326	1504
ENSG00000187961	120	153	180	236	255	357
ENSG00000187583	24	48	65	44	48	64
ENSG00000187642	4	9	16	14	16	16

```
metadata <- read.csv("GSE37704_metadata.csv")
metadata$id
```

```
[1] "SRR493366" "SRR493367" "SRR493368" "SRR493369" "SRR493370" "SRR493371"
```

```
dim(countData)
```

```
[1] 15975      6
```

Section 1: Setup DESeq Object

Now lets setup the DESeqDataSet object required for the DESeq() function and then run the DESeq pipeline.

```
dds <- DESeqDataSetFromMatrix(  
  countData = countData,  
  colData = colData,  
  design = ~condition  
)
```

Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in design formula are characters, converting to factors

```
dds <- DESeq(dds)
```

estimating size factors

estimating dispersions

gene-wise dispersion estimates

mean-dispersion relationship

final dispersion estimates

fitting model and testing

```
dds
```

```
class: DESeqDataSet  
dim: 15975 6  
metadata(1): version  
assays(4): counts mu H cooks  
rownames(15975): ENSG00000279457 ENSG00000187634 ... ENSG00000276345  
               ENSG00000271254  
rowData names(22): baseMean baseVar ... deviance maxCooks  
colnames(6): SRR493366 SRR493367 ... SRR493370 SRR493371  
colData names(2): condition sizeFactor
```

Run DESeq Analysis Pipeline

```
res = results(dds)
```

Q3. Call the `summary()` function on your results to get a sense of how many genes are up or down-regulated at the default 0.1 p-value cutoff.

```
summary(res)
```

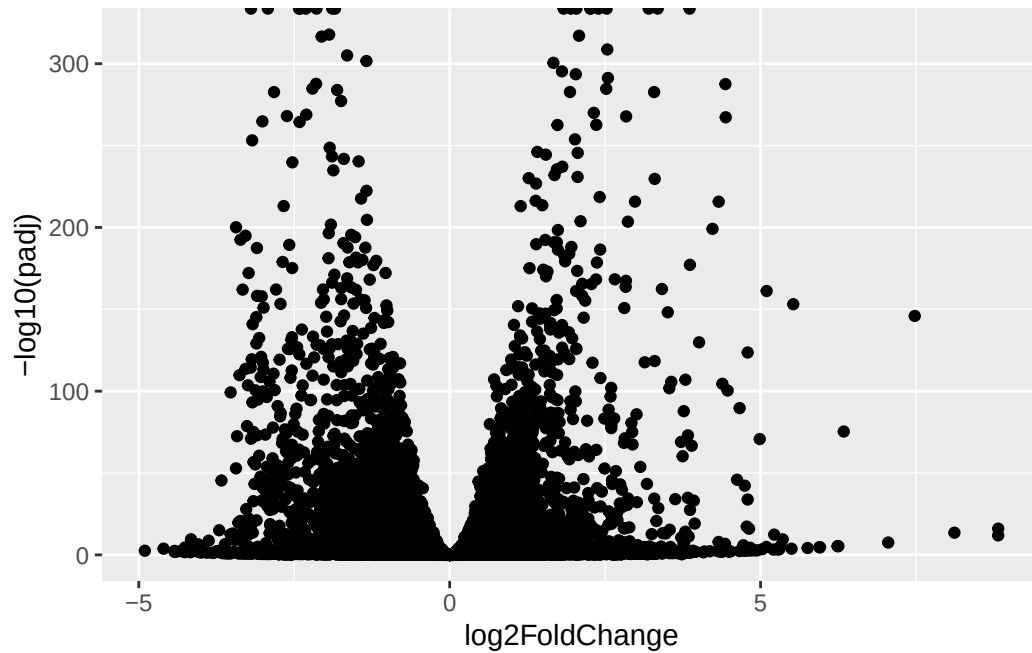
```
out of 15975 with nonzero total read count
adjusted p-value < 0.1
LFC > 0 (up)      : 4349, 27%
LFC < 0 (down)    : 4396, 28%
outliers [1]      : 0, 0%
low counts [2]    : 1237, 7.7%
(mean count < 0)
[1] see 'cooksCutoff' argument of ?results
[2] see 'independentFiltering' argument of ?results
```

Extract the Results

Big table with log2 fold changes and p-values:

```
ggplot(as.data.frame(res)) +
  aes(x = log2FoldChange, y = -log10(padj)) +
  geom_point()
```

Warning: Removed 1237 rows containing missing values or values outside the scale range (``geom_point()``).

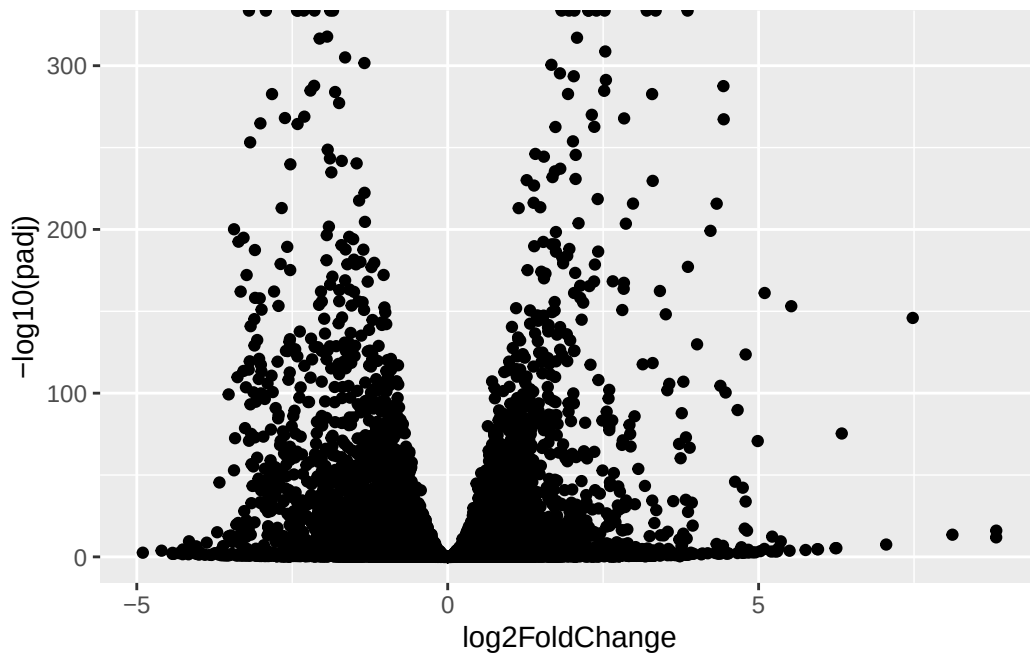


Data Visualization

Volcano Plot

```
ggplot(as.data.frame(res)) +  
  aes(x = log2FoldChange, y = -log10(padj)) +  
  geom_point()
```

Warning: Removed 1237 rows containing missing values or values outside the scale range (`geom\_point()`).

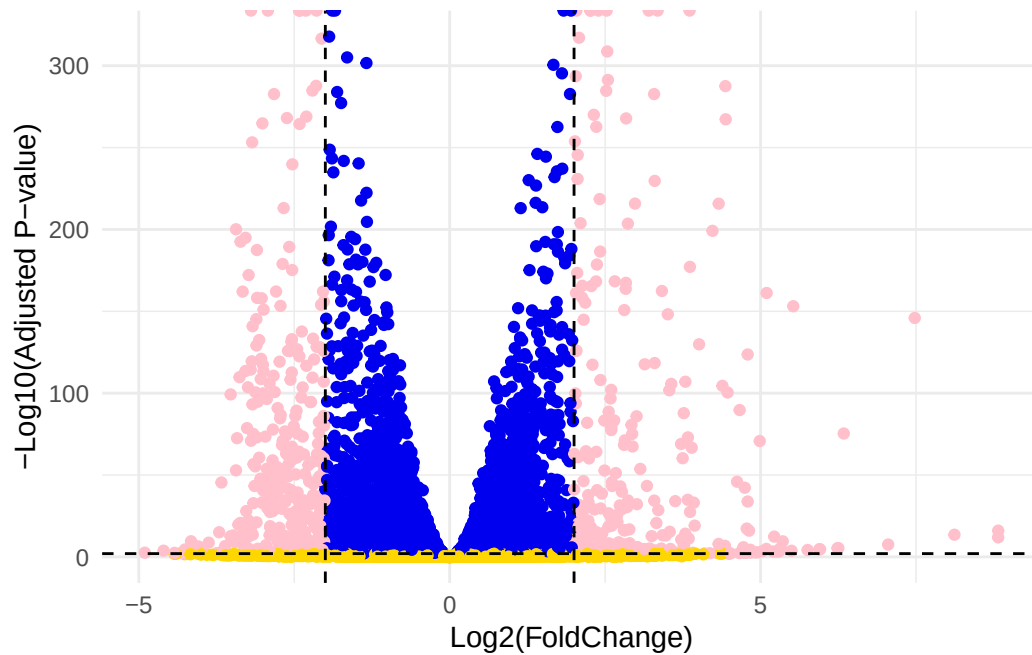


Q4. Improve this plot by completing the below code, which adds color, axis labels and cutoff lines:

```
mycols <- rep("blue2", nrow(res))
mycols[abs(res$log2FoldChange) > 2] <- "pink"
mycols[res$padj > 0.01 | is.na(res$padj)] <- "gold"

ggplot(as.data.frame(res)) +
  aes(x = log2FoldChange, y = -log10(padj)) +
  theme_minimal() +
  geom_point(col = mycols) +
  xlab("Log2(FoldChange)") +
  ylab("-Log10(Adjusted P-value)") +
  geom_vline(xintercept = c(-2, 2), linetype = "dashed", col="black") +
  geom_hline(yintercept = -log10(0.01), linetype = "dashed", col="black")
```

Warning: Removed 1237 rows containing missing values or values outside the scale range (`geom\_point()`).



Add Annotation Data

Q5. Use the `mapIds()` function multiple times to add SYMBOL, ENTREZID and GENENAME annotation to our results by completing the code below.

Add gene symbol and entrez IDs

```
library(AnnotationDbi)
library(org.Hs.eg.db)

res$symbol <- mapIds(
  org.Hs.eg.db,
  keys = row.names(res),
  keytype = "ENSEMBL",
  column = "SYMBOL",
  multiVals = "first"
)

res$entrez <- mapIds(
  org.Hs.eg.db,
  keys = row.names(res),
  keytype = "ENSEMBL",
```



```

    column = "ENTREZID",
    multiVals = "first"
)

res$name <- mapIds(
  org.Hs.eg.db,
  keys = row.names(res),
  keytype = "ENSEMBL",
  column = "GENENAME",
  multiVals = "first"
)

head(res, 10)

```

log2 fold change (MLE): condition hoxa1 kd vs control sirna

Wald test p-value: condition hoxa1 kd vs control sirna

DataFrame with 10 rows and 9 columns

	baseMean	log2FoldChange	lfcSE	stat	pvalue
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
ENSG00000279457	29.913579	0.1792571	0.3248215	0.551863	5.81042e-01
ENSG00000187634	183.229650	0.4264571	0.1402658	3.040350	2.36304e-03
ENSG00000188976	1651.188076	-0.6927205	0.0548465	-12.630156	1.43993e-36
ENSG00000187961	209.637938	0.7297556	0.1318599	5.534326	3.12428e-08
ENSG00000187583	47.255123	0.0405765	0.2718928	0.149237	8.81366e-01
ENSG00000187642	11.979750	0.5428105	0.5215598	1.040744	2.97994e-01
ENSG00000188290	108.922128	2.0570638	0.1969053	10.446970	1.51281e-25
ENSG00000187608	350.716868	0.2573837	0.1027266	2.505522	1.22271e-02
ENSG00000188157	9128.439422	0.3899088	0.0467164	8.346302	7.04333e-17
ENSG00000237330	0.158192	0.7859552	4.0804729	0.192614	8.47261e-01
	padj	symbol	entrez	name	
	<numeric>	<character>	<character>	<character>	
ENSG00000279457	6.86555e-01	NA	NA	NA	
ENSG00000187634	5.15718e-03	SAMD11	148398	sterile alpha motif ..	
ENSG00000188976	1.76553e-35	NOC2L	26155	NOC2 like nucleolar ..	
ENSG00000187961	1.13413e-07	KLHL17	339451	kelch like family me..	
ENSG00000187583	9.19031e-01	PLEKHN1	84069	pleckstrin homology ..	
ENSG00000187642	4.03379e-01	PERM1	84808	PPARGC1 and ESRR ind..	
ENSG00000188290	1.30538e-24	HES4	57801	hes family bHLH tran..	
ENSG00000187608	2.37452e-02	ISG15	9636	ISG15 ubiquitin like..	
ENSG00000188157	4.21970e-16	AGR1	375790	agrin	
ENSG00000237330	NA	RNF223	401934	ring finger protein ..	

Q6. Finally for this section let's reorder these results by adjusted p-value and save them to a CSV file in your current project directory.

```
res <- res[order(res$pvalue), ]
write.csv(res, file = "deseq_results.csv")
```

Section 2: Pathway Analysis

```
library(pathview)
library(gage)
library(gageData)

data(kegg.sets.hs)
data(sigmet.idx.hs)

# Focus on signaling and metabolic pathways only
kegg.sets.hs = kegg.sets.hs[sigmet.idx.hs]

# Examine the first 3 pathways
head(kegg.sets.hs, 3)
```

```
$`hsa00232 Caffeine metabolism`
```

```
[1] "10" "1544" "1548" "1549" "1553" "7498" "9"
```

```
$`hsa00983 Drug metabolism - other enzymes`
```

```
[1] "10" "1066" "10720" "10941" "151531" "1548" "1549" "1551"
[9] "1553" "1576" "1577" "1806" "1807" "1890" "221223" "2990"
[17] "3251" "3614" "3615" "3704" "51733" "54490" "54575" "54576"
[25] "54577" "54578" "54579" "54600" "54657" "54658" "54659" "54963"
[33] "574537" "64816" "7083" "7084" "7172" "7363" "7364" "7365"
[41] "7366" "7367" "7371" "7372" "7378" "7498" "79799" "83549"
[49] "8824" "8833" "9" "978"
```

```
$`hsa00230 Purine metabolism`
```

```
[1] "100" "10201" "10606" "10621" "10622" "10623" "107" "10714"
[9] "108" "10846" "109" "111" "11128" "11164" "112" "113"
[17] "114" "115" "122481" "122622" "124583" "132" "158" "159"
[25] "1633" "171568" "1716" "196883" "203" "204" "205" "221823"
[33] "2272" "22978" "23649" "246721" "25885" "2618" "26289" "270"
[41] "271" "27115" "272" "2766" "2977" "2982" "2983" "2984"
```

```

[49] "2986"    "2987"    "29922"   "3000"    "30833"   "30834"   "318"     "3251"
[57] "353"     "3614"    "3615"    "3704"    "377841"  "471"     "4830"    "4831"
[65] "4832"    "4833"    "4860"    "4881"    "4882"    "4907"    "50484"   "50940"
[73] "51082"   "51251"   "51292"   "5136"    "5137"    "5138"    "5139"    "5140"
[81] "5141"    "5142"    "5143"    "5144"    "5145"    "5146"    "5147"    "5148"
[89] "5149"    "5150"    "5151"    "5152"    "5153"    "5158"    "5167"    "5169"
[97] "51728"   "5198"    "5236"    "5313"    "5315"    "53343"   "54107"   "5422"
[105] "5424"    "5425"    "5426"    "5427"    "5430"    "5431"    "5432"    "5433"
[113] "5434"    "5435"    "5436"    "5437"    "5438"    "5439"    "5440"    "5441"
[121] "5471"    "548644"  "55276"   "5557"    "5558"    "55703"   "55811"   "55821"
[129] "5631"    "5634"    "56655"   "56953"   "56985"   "57804"   "58497"   "6240"
[137] "6241"    "64425"   "646625"  "654364"  "661"     "7498"    "8382"    "84172"
[145] "84265"   "84284"   "84618"   "8622"    "8654"    "87178"   "8833"    "9060"
[153] "9061"    "93034"   "953"     "9533"    "954"     "955"     "956"     "957"
[161] "9583"    "9615"

```

The main `gage()` function requires a named vector of fold changes, where the names of the values are the Entrez gene IDs.

Note: We used the `mapIDs()` function above to obtain Entrez gene IDs (stored in `res$entrez`) and we have the fold change results from DESeq2 analysis (stored in `res$log2FoldChange`).

```

foldchanges = res$log2FoldChange
names(foldchanges) = res$entrez
head(foldchanges)

```

```

      1266      54855      1465      2034      2150      6659
-2.422719  3.201955 -2.313738 -1.888019  3.344508  2.392288

```

Let's run the `gage` analysis

```

# Get the results
keggres = gage(foldchanges, gsets=kegg.sets.hs)

```

```

attributes(keggres)

```

```

$names
[1] "greater" "less"    "stats"

```

```
# look at the first few down, less, pathways
head(keggres$less)
```

	p.geomean	stat.mean	p.val
hsa04110 Cell cycle	8.995727e-06	-4.378644	8.995727e-06
hsa03030 DNA replication	9.424076e-05	-3.951803	9.424076e-05
hsa03013 RNA transport	1.375901e-03	-3.028500	1.375901e-03
hsa03440 Homologous recombination	3.066756e-03	-2.852899	3.066756e-03
hsa04114 Oocyte meiosis	3.784520e-03	-2.698128	3.784520e-03
hsa00010 Glycolysis / Gluconeogenesis	8.961413e-03	-2.405398	8.961413e-03

	q.val	set.size	exp1
hsa04110 Cell cycle	0.001448312	121	8.995727e-06
hsa03030 DNA replication	0.007586381	36	9.424076e-05
hsa03013 RNA transport	0.073840037	144	1.375901e-03
hsa03440 Homologous recombination	0.121861535	28	3.066756e-03
hsa04114 Oocyte meiosis	0.121861535	102	3.784520e-03
hsa00010 Glycolysis / Gluconeogenesis	0.212222694	53	8.961413e-03

Now, let's try out the `pathview()` function from the `pathview` package to make a pathway plot with our RNA-Seq expression results shown in color.

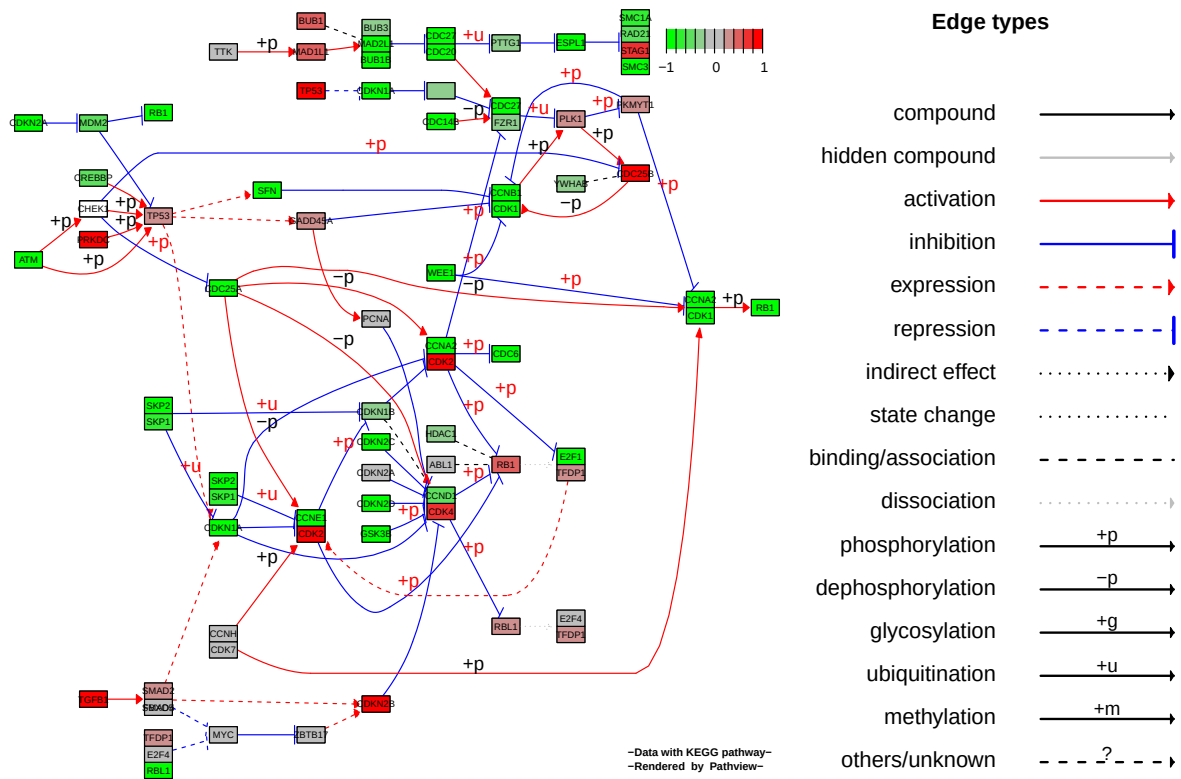
```
pathview(gene.data=foldchanges, pathway.id="hsa04110")
```

```
'select()' returned 1:1 mapping between keys and columns
```

```
Info: Working in directory /Users/melissabui/BIMM143/class14
```

```
Info: Writing image file hsa04110.pathview.png
```


Info: Writing image file hsa04110.pathview.pdf



Now, let's process our results a bit more to automatically pull out the top 5 upregulated pathways, then further process that just to get the pathway IDs needed by the `pathview()` function. We'll use these KEGG pathway IDs for pathview plotting below.

```
## focus on top 5 upregulated pathways here for demo purposes only
keggrespathways <- rownames(keggres$greater)[1:5]

# extract the 8 character long IDs part of each string
keggresids = substr(keggrespathways, start=1, stop=8)
keggresids
```

```
[1] "hsa04640" "hsa04630" "hsa00140" "hsa04142" "hsa04330"
```

Finally, let's pass these IDs in `keggresids` to the `pathview()` function to draw plots for all the top 5 pathways.

```
pathview(gene.data=foldchanges, pathway.id=keggresids, species="hsa")
```

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/melissabui/BIMM143/class14

Info: Writing image file hsa04640.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/melissabui/BIMM143/class14

Info: Writing image file hsa04630.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/melissabui/BIMM143/class14

Info: Writing image file hsa00140.pathview.png

'select()' returned 1:1 mapping between keys and columns

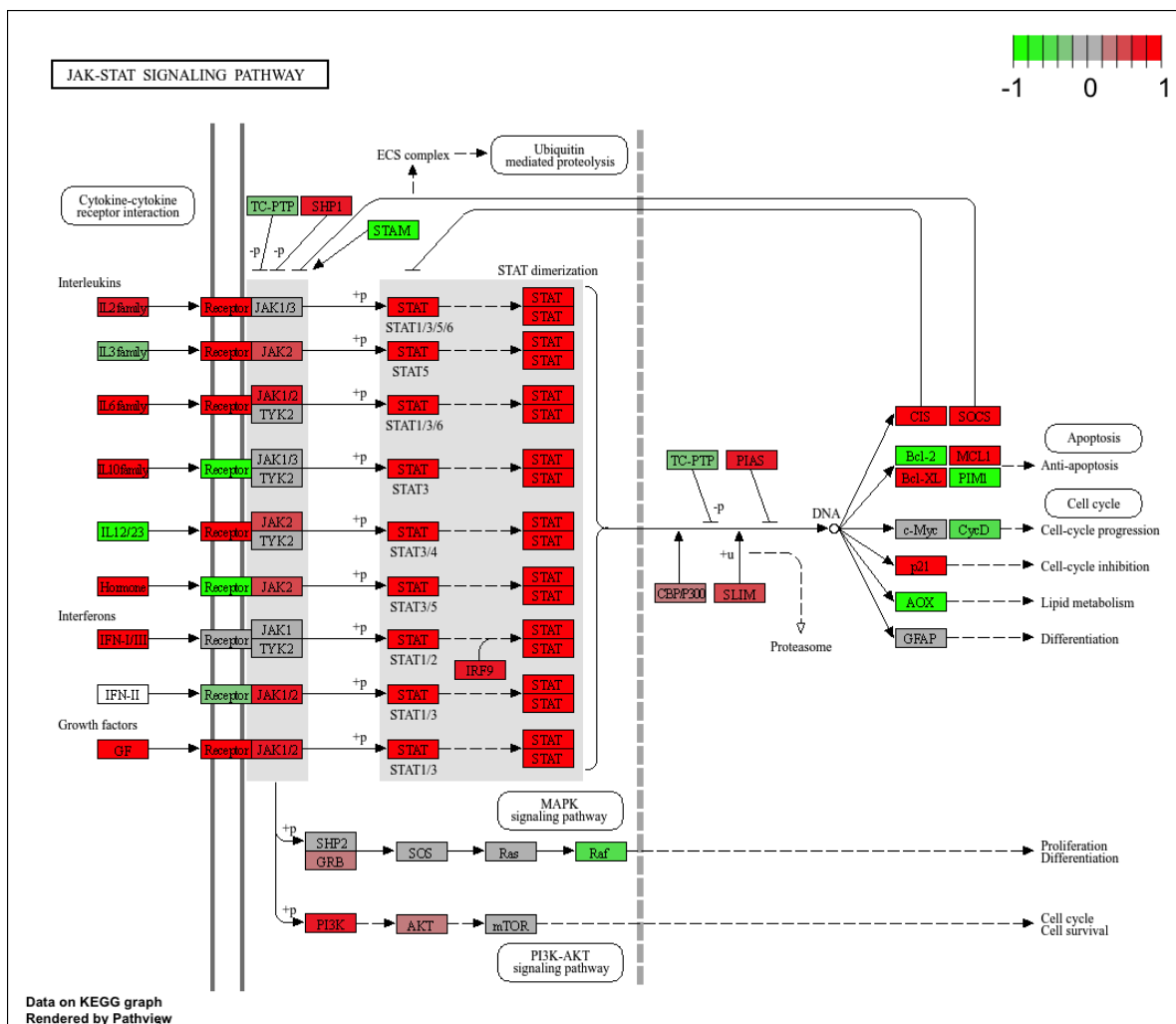
Info: Working in directory /Users/melissabui/BIMM143/class14

Info: Writing image file hsa04142.pathview.png

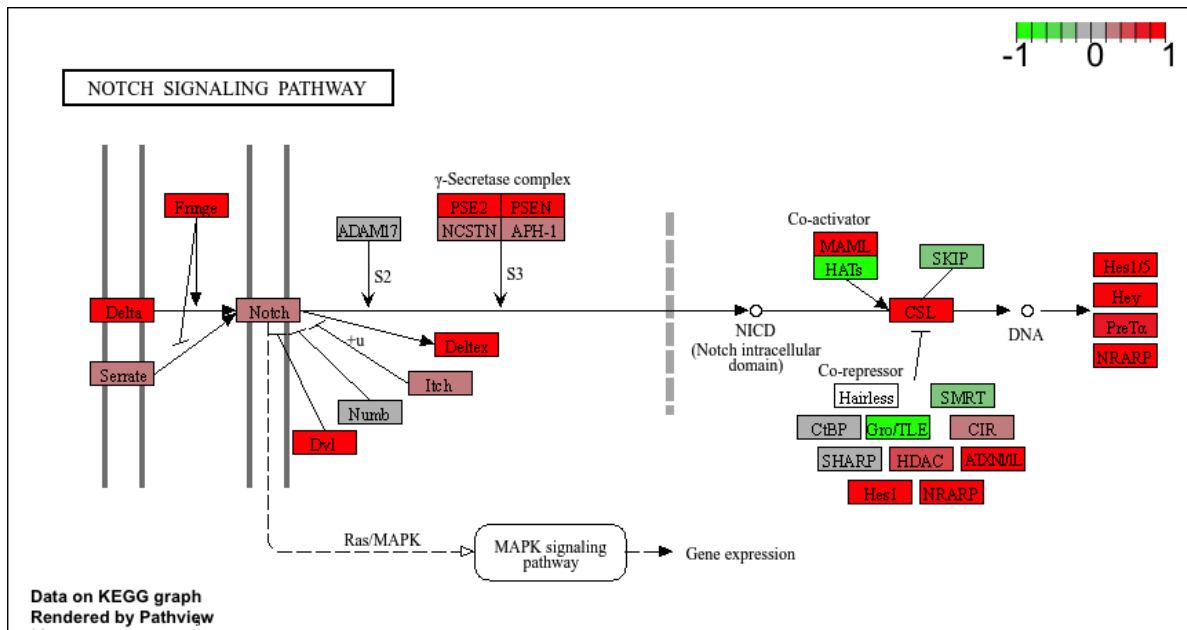
'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/melissabui/BIMM143/class14

Info: Writing image file hsa04330.pathview.png







Q7. Can you do the same procedure as above to plot the pathview figures for the top 5 down-regulated pathways?

```
# The top 5 down-regulated KEGG pathways
keggrespathways_down <- rownames(keggres$less)[1:5]

keggresids_down <- substr(keggrespathways_down,
                          start = 1,
                          stop = 8)

keggresids_down
```

```
[1] "hsa04110" "hsa03030" "hsa03013" "hsa03440" "hsa04114"
```

Draw pathview plots for the top 5 down-regulated pathways

```
pathview(gene.data = foldchanges,
         pathway.id = keggresids_down,
         species = "hsa")
```

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/melissabui/BIMM143/class14

Info: Writing image file hsa04110.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/melissabui/BIMM143/class14

Info: Writing image file hsa03030.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/melissabui/BIMM143/class14

Info: Writing image file hsa03013.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/melissabui/BIMM143/class14

Info: Writing image file hsa03440.pathview.png

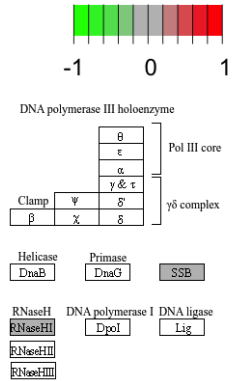
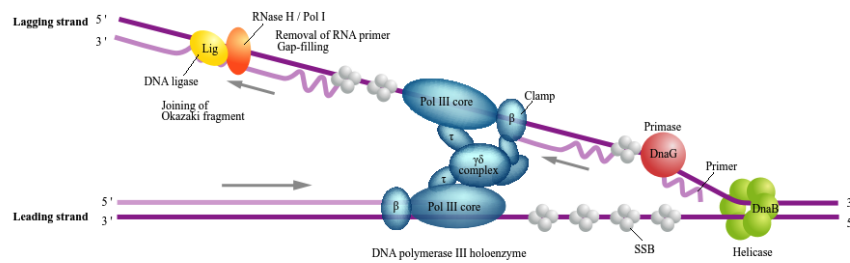
'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/melissabui/BIMM143/class14

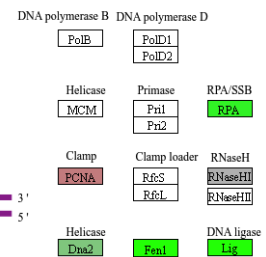
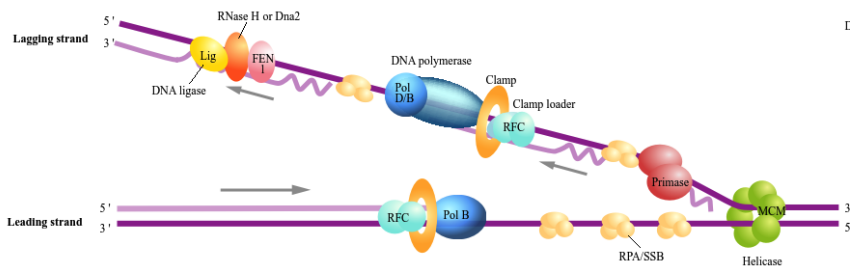
Info: Writing image file hsa04114.pathview.png

DNA REPLICATION

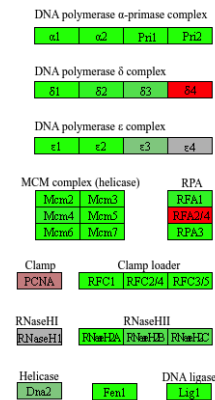
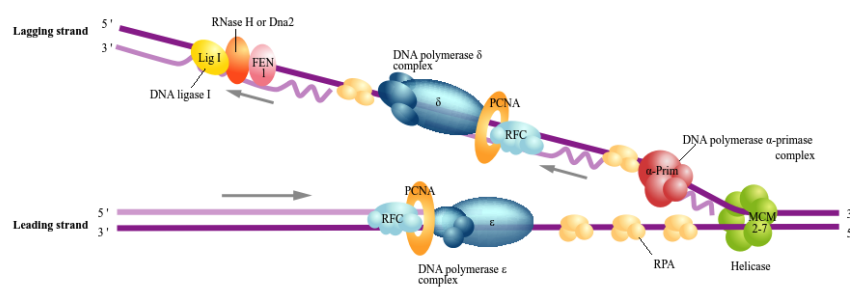
Replication complex (Bacteria)



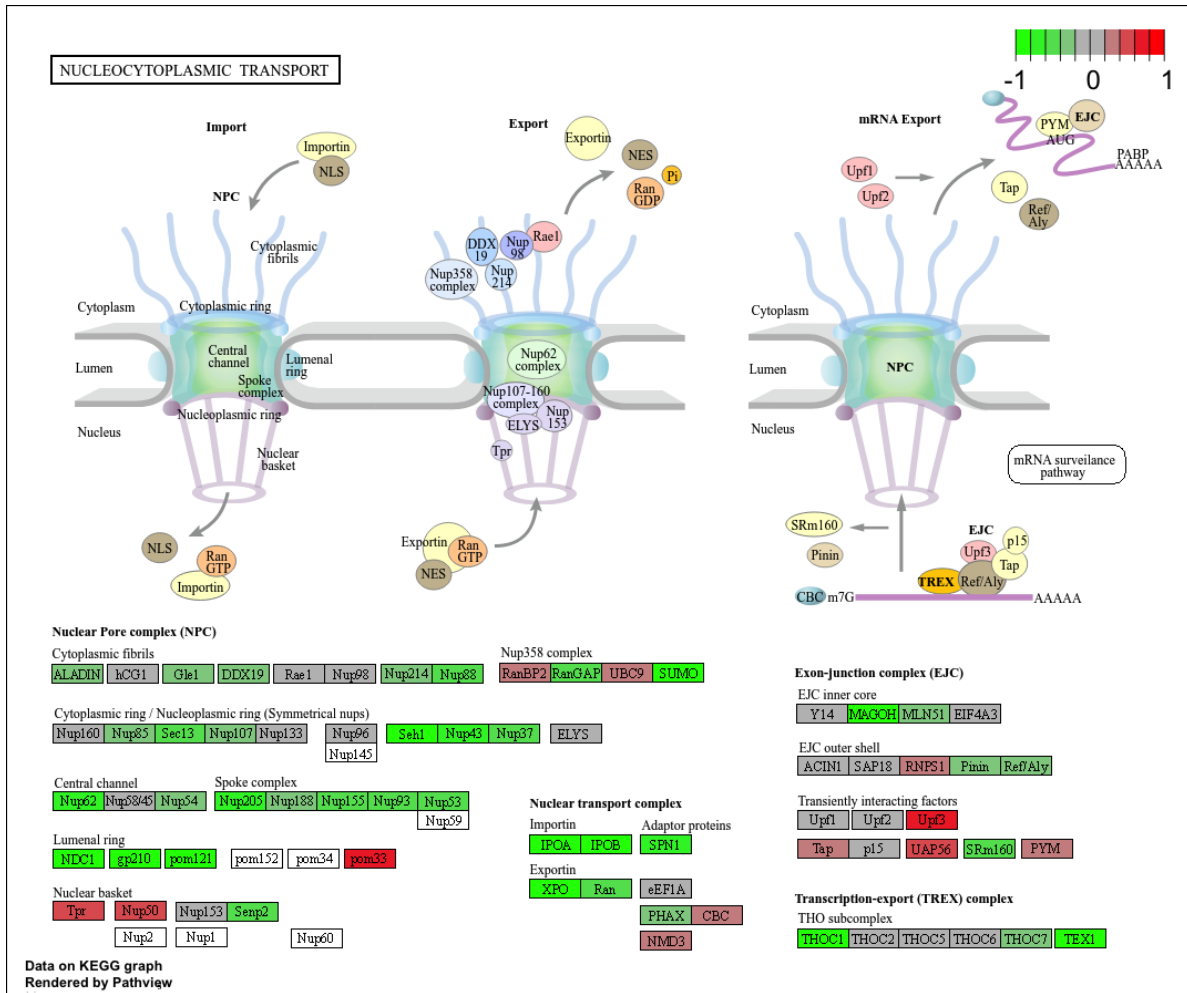
Replication complex (Archaea)

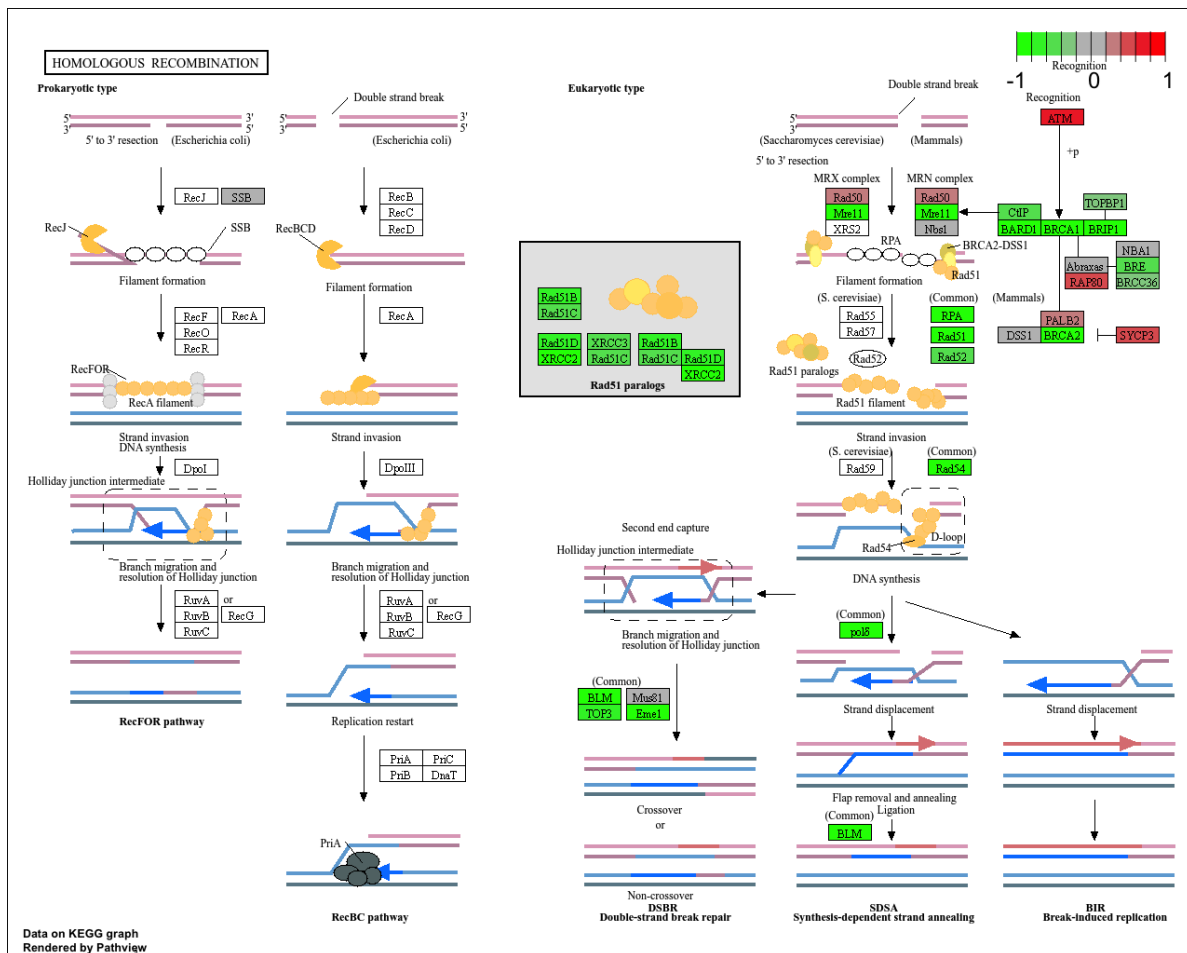


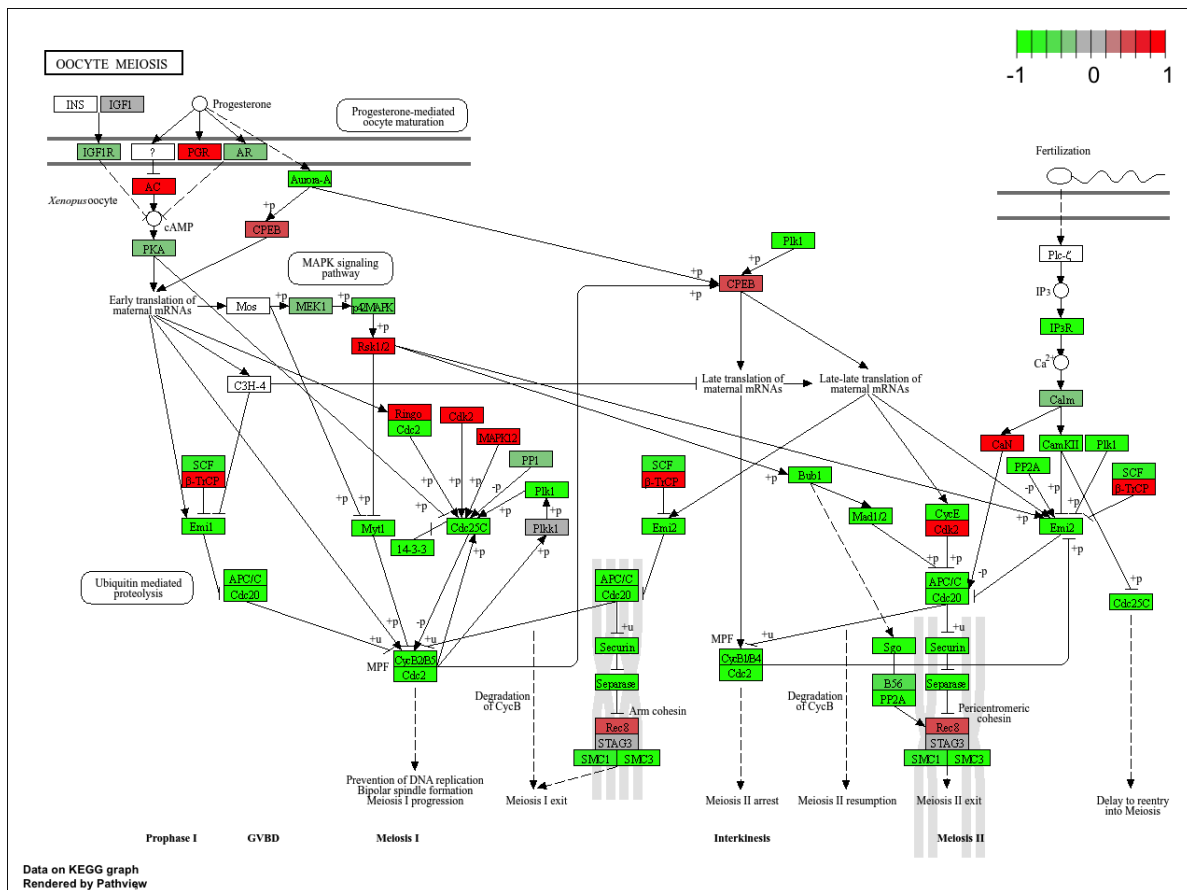
Replication complex (Eukaryotes)



Data on KEGG graph
Rendered by Pathview







Section 3: Gene Ontology (GO)

We can also do a similar procedure with gene ontology. Similar to above, `go.sets.hs` has all GO terms. `go.subs.hs` is a named list containing indexes for the BP, CC, and MF ontologies. Let's focus on BP (a.k.a Biological Process) here.

```
data(go.sets.hs)
data(go.subs.hs)

# focus on Biological Process subset of GO
gobpsets = go.sets.hs[go.subs.hs$BP]

gobpres = gage(foldchanges, gsets=gobpsets)

lapply(gobpres, head)
```

\$greater

		p.geomean	stat.mean	p.val
G0:0007156	homophilic cell adhesion	8.519724e-05	3.824205	8.519724e-05
G0:0002009	morphogenesis of an epithelium	1.396681e-04	3.653886	1.396681e-04
G0:0048729	tissue morphogenesis	1.432451e-04	3.643242	1.432451e-04
G0:0007610	behavior	1.925222e-04	3.565432	1.925222e-04
G0:0060562	epithelial tube morphogenesis	5.932837e-04	3.261376	5.932837e-04
G0:0035295	tube development	5.953254e-04	3.253665	5.953254e-04
		q.val	set.size	exp1
G0:0007156	homophilic cell adhesion	0.1951953	113	8.519724e-05
G0:0002009	morphogenesis of an epithelium	0.1951953	339	1.396681e-04
G0:0048729	tissue morphogenesis	0.1951953	424	1.432451e-04
G0:0007610	behavior	0.1967577	426	1.925222e-04
G0:0060562	epithelial tube morphogenesis	0.3565320	257	5.932837e-04
G0:0035295	tube development	0.3565320	391	5.953254e-04

\$less

		p.geomean	stat.mean	p.val
G0:0048285	organelle fission	1.536227e-15	-8.063910	1.536227e-15
G0:0000280	nuclear division	4.286961e-15	-7.939217	4.286961e-15
G0:0007067	mitosis	4.286961e-15	-7.939217	4.286961e-15
G0:0000087	M phase of mitotic cell cycle	1.169934e-14	-7.797496	1.169934e-14
G0:0007059	chromosome segregation	2.028624e-11	-6.878340	2.028624e-11
G0:0000236	mitotic prometaphase	1.729553e-10	-6.695966	1.729553e-10
		q.val	set.size	exp1
G0:0048285	organelle fission	5.841698e-12	376	1.536227e-15
G0:0000280	nuclear division	5.841698e-12	352	4.286961e-15
G0:0007067	mitosis	5.841698e-12	352	4.286961e-15
G0:0000087	M phase of mitotic cell cycle	1.195672e-11	362	1.169934e-14
G0:0007059	chromosome segregation	1.658603e-08	142	2.028624e-11
G0:0000236	mitotic prometaphase	1.178402e-07	84	1.729553e-10

\$stats

		stat.mean	exp1
G0:0007156	homophilic cell adhesion	3.824205	3.824205
G0:0002009	morphogenesis of an epithelium	3.653886	3.653886
G0:0048729	tissue morphogenesis	3.643242	3.643242
G0:0007610	behavior	3.565432	3.565432
G0:0060562	epithelial tube morphogenesis	3.261376	3.261376
G0:0035295	tube development	3.253665	3.253665

Section 4. Reactome Analysis

```
sig_genes <- res[res$padj <= 0.05 & !is.na(res$padj), "symbol"]  
print(paste("Total number of significant genes:", length(sig_genes)))
```

```
[1] "Total number of significant genes: 8147"
```

```
write.table(sig_genes, file="significant_genes.txt", row.names=FALSE, col.names=FALSE, quote=
```

Q8. What pathway has the most significant “Entities p-value”? Do the most significant pathways listed match your previous KEGG results? What factors could cause differences between the two methods?

The pathway with the most significant Entities p-value is the Cell Cycle pathway $p = 2.63 \times 10^{-5}$, followed closely by Cell Cycle, Mitotic, $p = 2.78 \times 10^{-5}$. Overall, the top Reactome pathways are consistent with the earlier KEGG analysis if KEGG also identified pathways related to the cell cycle, DNA replication, or mitosis, supporting the biological reliability of the results across databases. Differences between Reactome and KEGG pathway rankings and significance values may still occur since the two databases use different statistical approaches and pathway definitions.

Save Our Results

```
# Save differential expression results to a CSV file  
write.csv(res, "deseq_results.csv")  
  
# Save significant genes only  
write.csv(sig_genes, file = "significant_genes.csv")  
  
# Save fold change vector if needed  
write.csv(foldchanges, file = "foldchanges.csv")
```